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Mobile Computing Assignment

LAB 4_POST MANAGER

1. Dependencies Used

Flutter Lab 4 - Answers

1. Which dependencies did you use and why?

In this project, The main dependencies used is:

- http packages

The purpose of this package is to communicate with the Api (send and reciever data from server).

- It Allows us to perform:
  - GET: Fetch posts
  - POST: Create new post
  - PUT/PATCH: update post
  - DELETE: remove post.

Example

```
dependencies:
flutter:
  sdk: flutter
  http: ^0.13.6
```

Why we use http:

- It is simpler and easy for beginners
- Support All HTTP Methods
- Work well with REST Apis like JSONplaceholder

Example usage

```
Import 'package: http/http.dart' as http;
final response = await http.get(
 uri.parse ('https://Jsonplaceholder.typicode.com/posts');
);
```

Explanation

http.get() : fetch data from Api

uri.parse() : Convert url into usable format

await -> wait for the server response

2. Handling API Exceptions

2. How to handle Api Exceptions (Network error, invalid data)

Handling error is very Important to avoid app crashes

* Common Api errors:
  1. No internet Connection
  2. Server error (500, 404)
  3. Invalid data (Wrong JSON format)

* Solution: use try-catch

```
Future<list<post>> fetchposts () async {
try {
  final response = await http.get (
  Uri.parse ('https://Jsonplaceholder.typicode.com/posts'),
  );
  if (response.statuscode == 200) {
    return postfromJson (response.body);
  } else {
    Throw Exception ("failed to load posts");
  }
  } Catch (e) {
    Throw Exception ("Network error: $e");
  }}
```

Best practices:

- show Loading indicator
- show error message to user
- Avoid app crash.

UI Example:

```
if (snapshot.has_error) {
  return Center (child: Text ("error: ${snapshot.error}")),
}
```

3. Future Widget used and which UI depends on it

3. Explain the future Widget used and which UI depends on it.

A Future: represents a value that will be available later (after loading from API).

* Widget used: FutureBuilder
- It is used to build UI based on API data
- Automatically updates UI when data arrives.

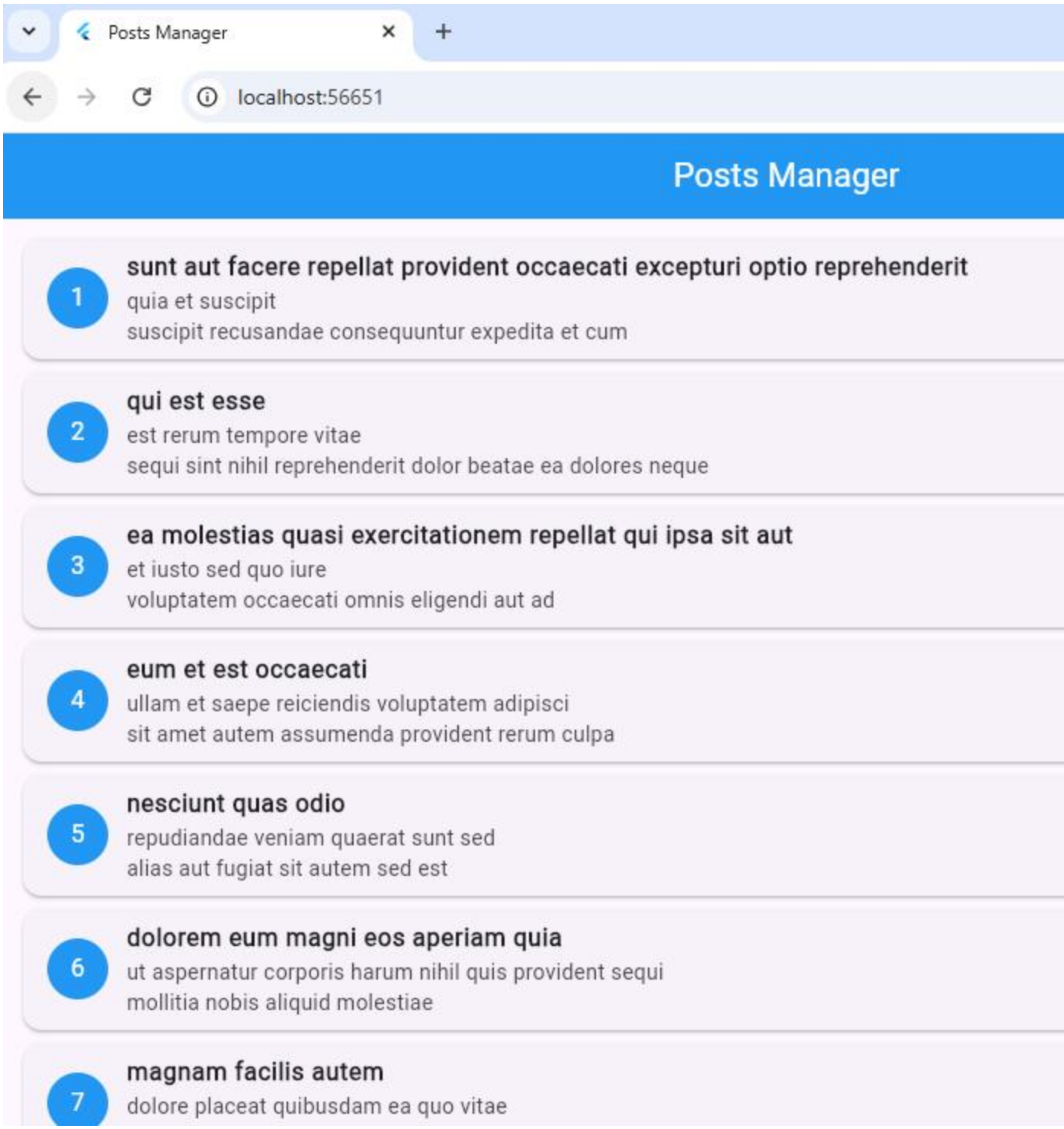
Example

```
FutureBuilder < list < post >> (
 Future : fetchPosts ( ),
 Builder : ( Content, Snapshot) {
 if (Snapshot. Connectionstate == Connectionstate. waiting) {
   return circularprogressIndicator ( );
 }else if (Snapshot . haserror) {
  return Text ("error : ${ Snapshot . error}");
 } else {
     return ListView . builder (
    itemCount : Snapshot . data ! . length,
    itemBuilder : (Content, index ) {
    return ListTile (
        title : Text (Snapshot . data ! [Index]. title),
      );
    };
    );
    }
    },
    )
```

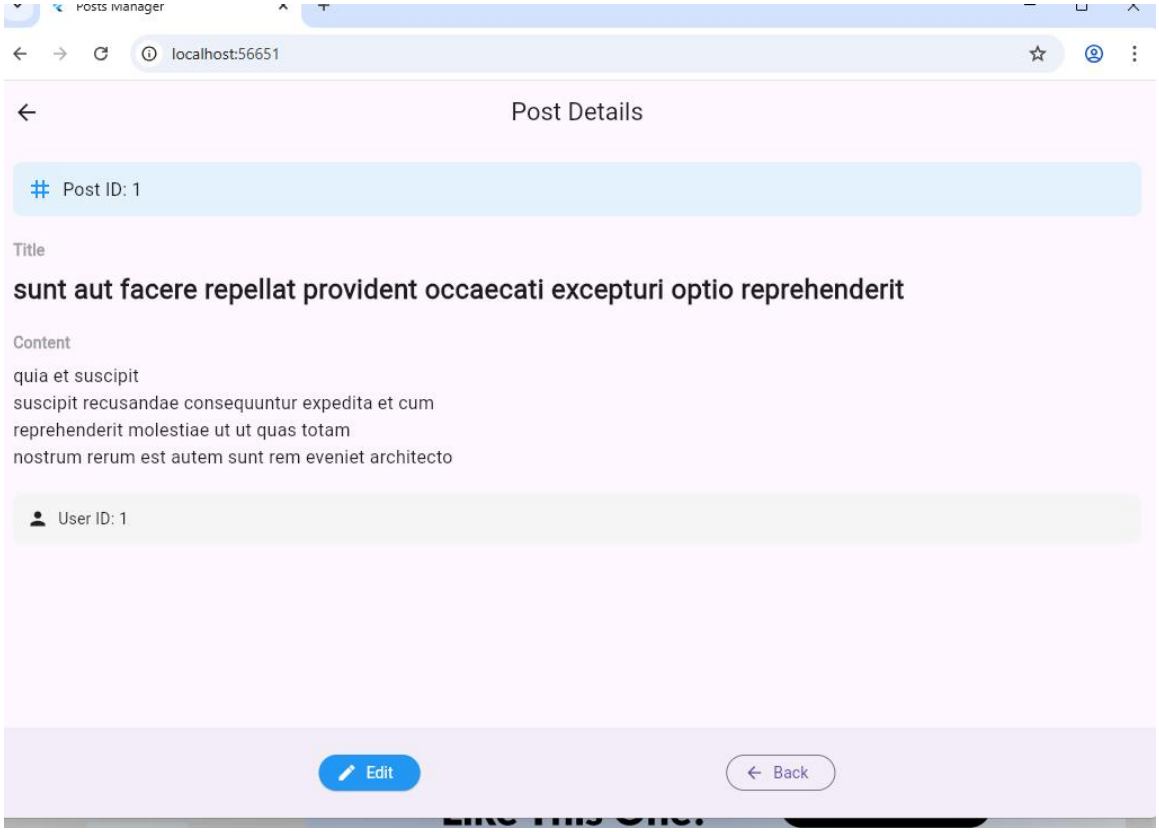
4. Features Implemented

4. ScreenShoot

- View all posts



- View post details


Posts Manager
localhost:56651
Post Details
Post ID: 1
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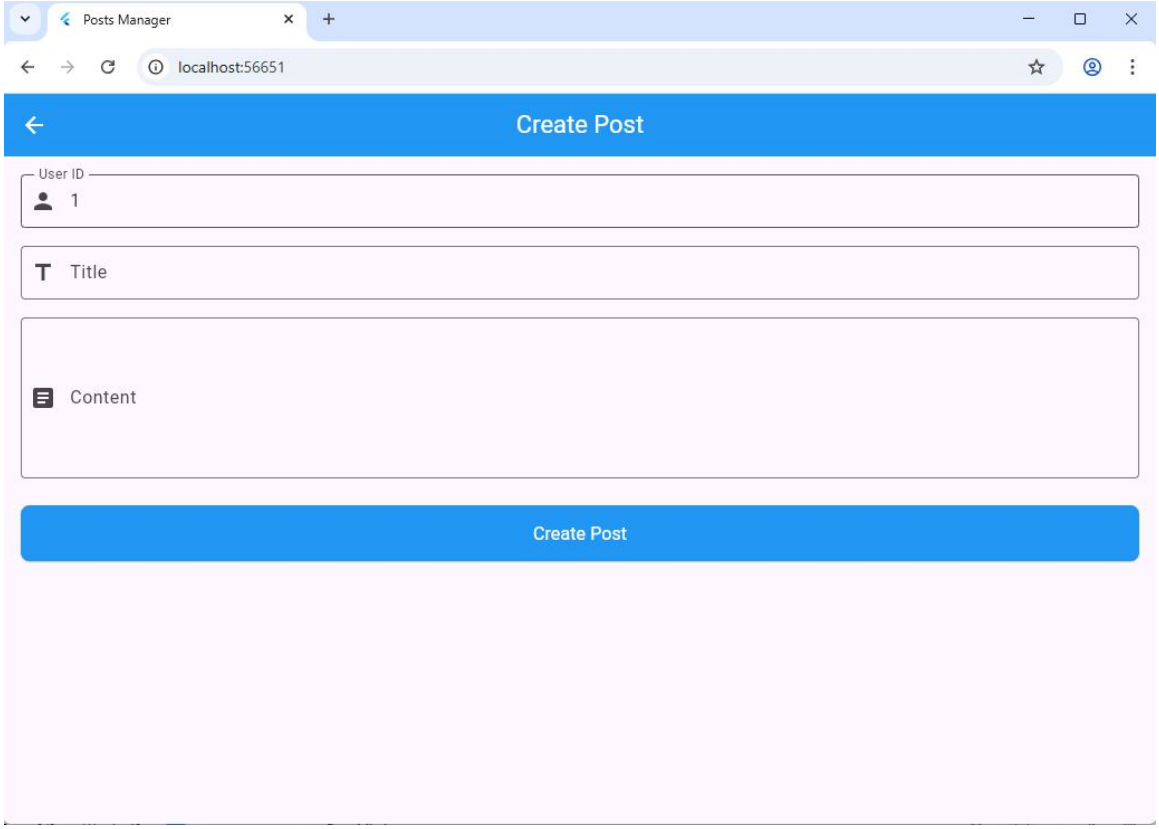
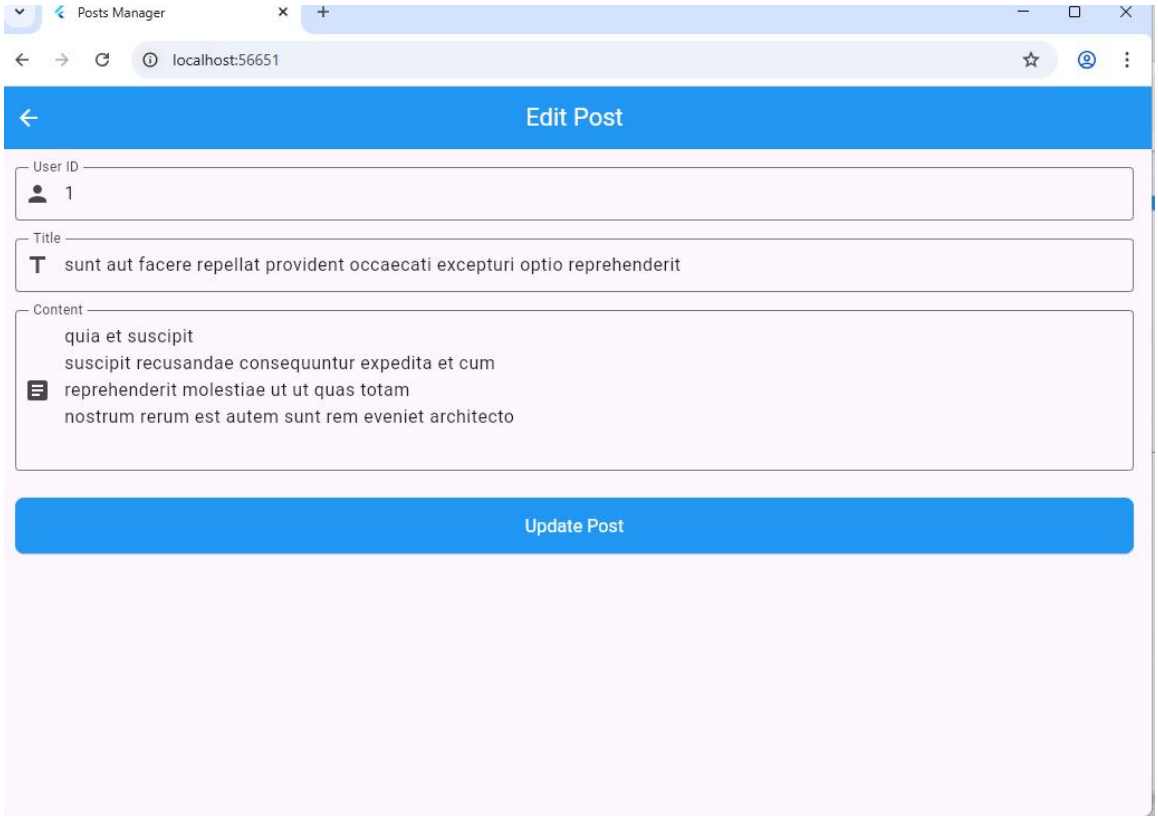
User ID: 1
Edit
Back

- Add new post


Posts Manager
localhost:56651
Create Post
User ID
1
Title
Content
Create Post

- Edit post


Posts Manager
localhost:56651
Edit Post
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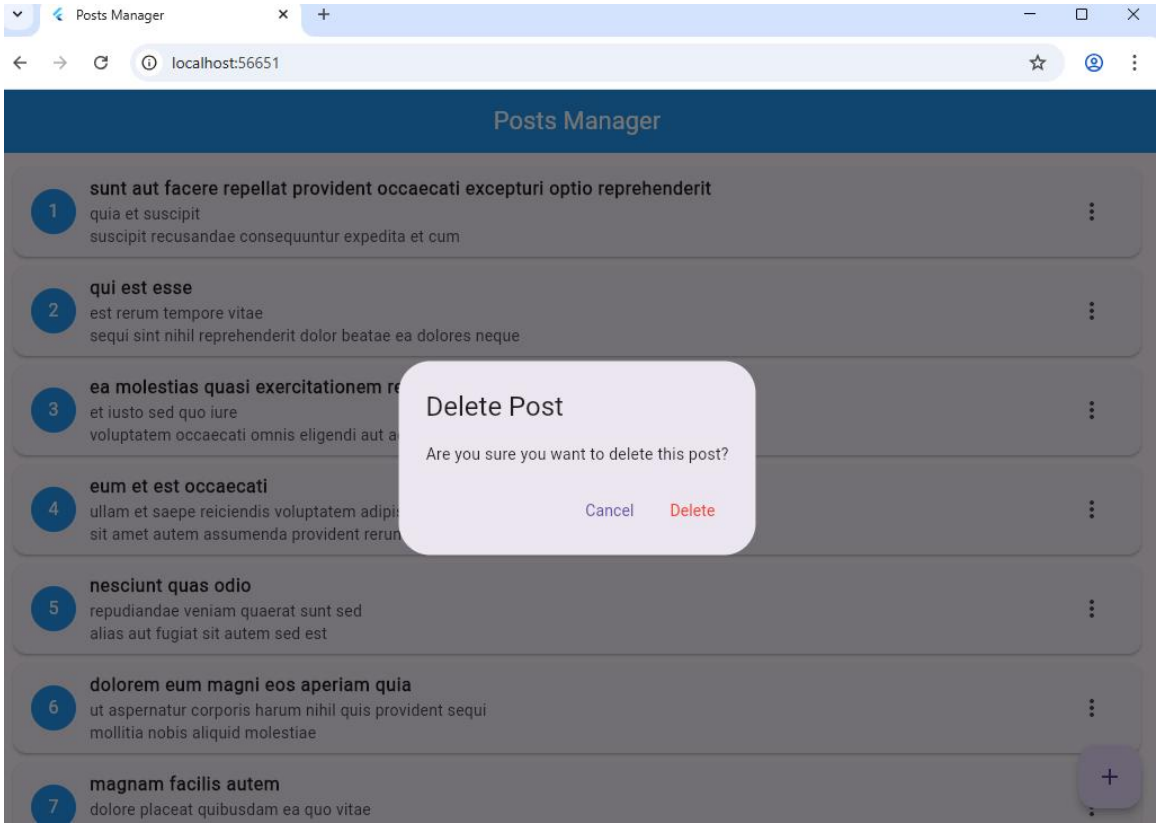
Update Post

- Delete post


Posts Manager
localhost:56651
Posts Manager
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7
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dolore placeat quibusdam ea quo vitae
Delete Post
Are you sure you want to delete this post?
Cancel
Delete